Predicting the effect and the mechanism of BRCA1 missense variants

Meet-EU course · one semester, October–January · multi-team · 4EU+ activity

1. Introduction

BRCA1 is one of the few genes where a variant interpretation changes what happens to a patient the same week: risk-reducing surgery, intensified screening, PARP-inhibitor eligibility and cascade testing in the family all hang on whether a variant is called pathogenic, benign — or left as a variant of uncertain significance (VUS). Decades of testing have produced thousands of BRCA1 missense variants that sit in that third category. Each of them is a small unsolved problem with a real person attached to it.

BRCA1 is also an unusually honest test of how good our computational biology really is. The protein is 1,863 residues long, but only two small parts of it are folded: the N-terminal RING domain, which forms an obligate heterodimer with BARD1, and the C-terminal tandem BRCT repeats, which read phosphorylated peptides on partner proteins. Between them lies more than a thousand residues of intrinsically disordered sequence, punctuated by a coiled-coil that binds PALB2 and by phosphorylation sites for ATM, ATR and CHK2. Structure-based predictors have almost nothing to say about that middle region, and multiplexed functional assays have barely touched it.

The design of the project follows the CAGI “Annotate All Missense” challenge, where teams predict pathogenicity for very large sets of variants and are scored later, as clinical and experimental annotations accumulate. We keep the prospective logic of that challenge but shrink the target to a single protein, so that **you can think about biology rather than about file formats**: every variant you predict on sits in a protein whose domains, partners and assays you can actually learn.

2. Research question

For every missense variant of BRCA1 that is currently classified in ClinVar as “uncertain significance” or “conflicting classifications” (4,606 variants in the ClinVar snapshot of 5 October 2026: 1,932 classified as uncertain significance and 2,674 as conflicting), each team answers two questions:

1. **Effect.** Is the variant damaging to BRCA1 function, and with what confidence? The answer is a calibrated probability, not a label — your program must tell how sure they are, variant by variant.
2. **Mechanism.** For variants predicted to be damaging, why? The answer is one of the mechanism classes in section 3, with the evidence that supports it. A prediction of “pathogenic” without an indication of what is broken is only half an answer.

The predictions are frozen and submitted on a fixed date. Part of the assessment happens straight away, on the data every team already has; the other part happens months later, when ClinVar reclassifies some of these variants and the teams can see how it did. That delayed feedback is deliberate: it is what doing science actually feels like.

3. A mechanism taxonomy to predict

The taxonomy is two-level on purpose. A team that can tell that a variant breaks a binding surface, but cannot say which one, annotates the parent class and says so; a team with structural evidence for a specific partner annotates the child class. Resolution is rewarded, guessing is not. “Damaging, mechanism unknown” is a legitimate answer and must be used rather than inventing a story.

Mechanism class	What is broken	Evidence a team can bring
Binding disruption (parent class)	A protein–protein or metal-binding surface is compromised, but the evidence does not say which	Residue on a known interface or metal site; loss of function with the fold intact
— zinc-site disruption	One of the cysteine/histidine ligands of the two RING zinc sites is lost	Metal-binding annotation; structure of the RING domain
— BARD1 interface	The RING–RING heterodimer with BARD1 is weakened	Interface in the BRCA1–BARD1 structure; E3 ligase activity assays

— BRCT phosphopeptide pocket	The pocket that reads phosphorylated ABRAXAS1, BRIP1 or CtIP is altered	BRCT–phosphopeptide structures; functional assays of the BRCT region
— PALB2 coiled-coil	The PALB2-binding coiled-coil (~res. 1,390–1,420) is disrupted	Position within the coiled-coil; conservation
Fold destabilisation	The RING or BRCT domain no longer folds or is destabilised	Predicted stability change; burial in the folded domain
Phosphorylation-site loss	An ATM, ATR or CHK2 phosphosite is destroyed (or a new one created)	Phosphosite annotation with functional evidence; site conservation
Disrupts splicing	The nucleotide change breaks splicing rather than the protein	Splice-effect predictions; position relative to the exon boundary
Damaging, mechanism unknown	The variant is predicted to matter, but no mechanism is supported	Say so explicitly, and state what evidence would be needed to resolve it
No effect	Tolerated substitution	Population frequency, tolerance across orthologues, functional assay in the normal range

4. Pipeline architecture (the shared chassis)

Every team can follow the same six-stage pipeline. What differs is the biology and the modelling choices each team puts into it — not the part of the protein it covers.

1. **Data curation.** Start from the frozen core dataset (section 5). Build the training set from variants that are NOT in the prediction set: ClinVar pathogenic/likely pathogenic and benign/likely benign, multiplexed assay scores from MaveDB, gnomAD frequencies. Document every filter.
2. **Feature extraction.** At minimum: evolutionary (alignment-based conservation, a protein language model), structural (RING and BRCT: ΔΔG, RSA, contacts; whole protein: AlphaFold pLDDT as a disorder proxy), interaction (distance to an annotated interface or PTM site) and RNA (splice-effect scores).
3. **Baselines.** Reproduce at least two published predictors on the training set and measure them honestly. Everything the team builds afterwards is judged against these numbers, not against zero.
4. **Model.** A classifier of the team's choosing. Report probabilities, calibrate them and check calibration on held-out data.
5. **Validation.** Each team designs and declares its own three-way split of the labelled data: training, validation for tuning hyperparameters, and a test partition touched once, at the end. How each team chooses to cut the data is itself a result, and the designs are compared at the final workshop.
6. **Prediction and mechanism assignment.** Apply the frozen pipeline to all 4,606 VUS and conflicting variants. For every variant predicted damaging, assign a mechanism class and a one-sentence justification. Write up ten of them properly as mechanism cards (section 7).

5. Datasets

Resource	What you get	Link
Frozen core dataset (distributed)	Three TSV files pulled from ClinVar on 5 October 2026 and frozen: brca1_clinvar_2026-10-05_predict.tsv (the 4,606 variants to predict), _reference_labels.tsv (696 confidently classified substitutions for calibration) and _all.tsv (the full 6,463-record pull). Each carries its query, transcript, build and column documentation in the header; README_brca1_clinvar_dataset.md explains what was excluded and why	*Distributed by the instructors; do not re-pull ClinVar during the project.*

ClinVar (BRCA1)	The live database behind the frozen files: classifications, review status, submitters, submission-level evidence. Use it to read the evidence for individual variants — not to rebuild the prediction set	ncbi.nlm.nih.gov/clinvar
MaveDB — BRCA1	Multiplexed assays of variant effect: saturation genome editing (Findlay 2018), RING-domain E3 ligase activity (Starita 2015), BRCT multiplexed DSB-repair assays	mavedb.org — BRCA1
gnomAD	Population allele frequencies; a variant seen repeatedly in healthy adults is evidence in itself	gnomad.broadinstitute.org
UniProt P38398	Domains, zinc-binding residues, phosphosites, interaction partners, isoforms	uniprot.org/P38398
PDB	Experimental structures: BRCA1–BARD1 RING heterodimer (1JM7), BRCT repeats alone and bound to phosphopeptides (1JNX, 1T29 and others)	rcsb.org
AlphaFold DB	Full-length model with per-residue pLDDT — in BRCA1, low pLDDT is itself informative	alphafold.ebi.ac.uk/entry/P38398
BRCA Exchange	Aggregated BRCA1/2 variant classifications across databases, including ENIGMA calls	brcaexchange.org
ENIGMA / ClinGen VCEP specifications	How experts actually weigh evidence for BRCA1 — the rulebook your predictions should be able to feed into	cspec.genome.network — BRCA1
ELM	Short linear motifs — the main handle on the disordered middle of the protein	elm.eu.org
dbNSFP / precomputed scores	Existing predictor scores for every possible missense variant; use as baselines, and beware of circularity	CAGI7 Annotate All Missense

> Circularity is the trap of this field. Most published predictors were trained on ClinVar labels; several MAVE datasets have already been used by expert panels to reclassify the very variants you might use as ground truth. Teams must state, explicitly and in writing, which of their features could have seen their evaluation labels, and what they did about it. Getting this wrong is the most common way a variant-effect paper dies in review.

6. The domains of BRCA1

Every team predicts every variant in the dataset, with the same deliverables and the same deadline.

- **The RING domain (res. ~1–110).** Two zinc sites; forms an obligate heterodimer with BARD1 and makes the pair an E3 ubiquitin ligase. The best-characterised region, structurally and functionally.
- **The disordered middle (res. ~110–1,640).** More than a thousand residues with no fixed structure, carrying the ATM/ATR/CHK2 phosphosites, nuclear localisation signals and short linear motifs. Most of the uncertain variants are here, and most tools have little to say about them.
- **The coiled-coil (res. ~1,390–1,420).** Binds PALB2 and through it BRCA2 — a short, structured island in the middle region, and the link into homologous recombination.
- **The tandem BRCT repeats (res. ~1,640–1,863).** A phosphopeptide-reading module that binds ABRAXAS1, BRIP1 and CtIP. Clinically dense and well covered by functional assays.

Nothing is withheld. Every team receives everything the instructors have — the full ClinVar extract, the confidently classified variants, the multiplexed assay data and the literature — and is expected to use all of it. Honesty about performance is established by how a model is validated and by whether another team can reproduce it, not by what was kept out of the team's hands.

7. Deliverables

- **Prediction file** (TSV, CAGI-style): variant in HGVS protein and genomic notation, probability of being damaging (0–1), standard deviation or confidence interval, class call (damaging / tolerated / unknown), mechanism class, and a free-text comment. Every variant gets a row; empty cells are not allowed.

- **Ten mechanism cards:** one page per variant for the ten highest-confidence damaging predictions — what the residue does, what the substitution should break, what evidence supports it, and what experiment would settle it.
- **Split policy file** (TSV plus half a page of rationale): every labelled variant the team used, assigned to train, validation or test, with the rule that produced the assignment. Filed with the pre-registration, before modelling, and unchanged thereafter.
- **Methods description** (3–5 pages): data, features, model, validation, calibration, and an explicit circularity statement.
- **Code repository:** runnable from raw data to prediction file, with a README and pinned dependencies. Another team must be able to reproduce your numbers.
- **Final presentation:** Brief oral summary of approach and results followed by poster session dedicated to more detailed discussions about chosen methods and gained insights.

8. Rules of engagement

The project is about reasoning with data, not about calling an API.

- **A submission may not be the output of a single third-party predictor.** Running one tool over the variant list and plotting the result is not a project. Published scores are welcome as features and required as baselines, but the team's model must combine at least two independent kinds of evidence — for example evolutionary and structural, or structural and RNA-level.
- **Uncertainty is mandatory.** Every prediction carries a calibrated probability and an uncertainty. “Unknown” is an acceptable class call when the evidence genuinely does not decide; a well-placed “unknown” scores better than a confident mistake.
- **Show your baselines.** Any claim of improvement is measured against at least two published predictors on the same held-out data, with the same metric. Try it!
- **Biology in the write-up.** A mechanism call must reference a residue, a structure, a motif or an assay. “The model assigned high importance to this feature” is not a mechanism. A mechanism label obtained by thresholding a predictor is not independent evidence.
- **Use of AI assistants is expected and must be declared,** with what was used and for what.

9. How the work is evaluated

This is not a competition. The project exists so that each team can take one real, unsolved research question the whole way through, and the assessment is built to reward that rather than to produce a winner.

The following aspects will however be considered:

Axis	What is measured	How
Discrimination	Does the model separate damaging from tolerated variants?	AUC and average precision on the 696 reference-labelled variants, with bootstrap confidence intervals
Coverage and abstention	Is the team honest about what it does not know?	Yield at a fixed precision: how many VUS could be moved with defensible evidence at, say, 95% precision
Mechanism plausibility	Are the mechanism calls specific and defensible?	Spatial-clustering permutation test, consistency with structures and assays, panel review of the mechanism cards
Reproducibility	Can someone else obtain the same file?	The repository runs from raw data to prediction file and gives the same numbers on another machine

Benchmarks and controls

- **Everything is released.** There is no instructor-held hold-out. The 696 confidently classified BRCA1 missense variants ship with the dataset and every team may use all of them. In exchange, each team writes down how it split the data before it starts modelling, reports its results against that split, and states in the methods which of its features could have seen its evaluation labels.
- **Reference-labelled variants.** 232 pathogenic/likely pathogenic and 464 benign/likely benign, expert-panel reviewed where the review status says so. Clinically meaningful, and the same kind of label the prediction set will eventually carry — but only 696 of them, concentrated in the RING and BRCT regions, so any confidence interval computed here is wide.

- **The multiplexed assay data is not a benchmark either.** It is open, and several published predictors were trained on it, so scoring against it measures memorisation as much as prediction. The assays stay in the toolbox as training data and as evidence for mechanism calls.
- **Split design is itself assessed.** Because every team cuts the data its own way, the headline numbers are not comparable across teams and are not treated as if they were. Two things are compared instead. First, the designs: random, position-grouped, domain-held-out, stratified by ClinVar review status, or chronological by date of last evaluation — the last of these imitates the real prospective problem and is possible because the dataset carries that date. Second, the optimism gap: the instructors re-score every team under one standard protocol and report the difference from the number the team reported itself. A large gap is not a failure to be punished; it is the most informative thing the cohort will produce about how splits flatter models, and the team that explains its own gap well has learned the lesson.
- **Negative controls that must fail.** Retrain on shuffled labels: performance has to collapse. If a model "works" on shuffled labels, there is a leak, and finding it is worth more marks than a high AUC.
- **Transfer test (optional).** Apply the pipeline unchanged to another protein with public functional data — BRCA2, PALB2, RAD51C, PTEN, TP53. A pipeline that survives transfer learned biology; one that collapses learned BRCA1 trivia. A joint exercise, not a contest.

The delayed check

Six to twelve months after the submission freeze, the instructors re-query ClinVar and report what happened to the predicted variants. Set expectations honestly: of roughly 4,600 BRCA1 missense variants of uncertain or conflicting status, a six-month window typically yields tens of genuine reclassifications, not hundreds, and expert panels work in batches — a new saturation-editing paper or an ENIGMA release can move hundreds of variants in one month, while a quiet half-year moves almost none. The result is circulated as a story, never as a grade: the sample is too small and too non-random to rank anyone, and treating it as a score would teach the wrong lesson about statistics.

10. Meetings and deadlines

Three fixed meetings anchor the project. The dates below do not move; everything between them is working time.

Meeting	When	Where	What happens
Kick-off	Friday 16 October 2026, 14:00	Zoom	The problem, the frozen dataset, the mechanism taxonomy and the rules of engagement. Teams are formed and the data is distributed.
Mid-term checkpoint	Friday 4 December 2026, 14:00	Zoom	Each team presents its reproduced baselines and its validation design. Peer review round: teams critique each other's designs while changing them is still cheap.
Final workshop	Friday 29 January 2027, full day	In person, Warsaw	Results presented: a brief oral summary from each team, then a poster session for the detail. Joint discussion of how teams split their data and where their predictions disagree; retrospective.

Both online meetings use the same Zoom room: cesnet.zoom.us/j/96292456885 (meeting ID 962 9245 6885; the passcode is embedded in the link).

The submission freeze is Tuesday 26 January 2027, three days before the workshop. Nothing is changed after that date: the gap exists so that the instructors can prepare the joint comparison shown at the workshop.

11. Practicalities

The project is run jointly by six universities. The scientific task, the dataset and the deadlines in section 10 are the same everywhere; everything administrative is local.

- **Team size.** Three to five people. Every team predicts every variant in the dataset.
- **Marks.** Your grade is set by the instructor at your own university, who will tell you how it is weighted. This document defines the work, not the marking scheme.
- **Poster.** Size and format for the final workshop will be announced closer to the date.
- **Questions.** Ask the instructor at your own university first — they know what your institution expects:

- **Prague —** Marian Novotný, marian@natur.cuni.cz
- **Warsaw —** Joanna Sułkowska, j.sulkowska@cent.uw.edu.pl
- **Copenhagen —** Amelie Stein, amelie.stein@bio.ku.dk
- **Paris —** Elodie Laine, elodie.laine@sorbonne-universite.fr
- **Milan —** Giulio Vistoli, giulio.vistoli@unimi.it
- **Heidelberg —** Carl Herrmann, carl.herrmann@bioquant.uni-heidelberg.de

12. Suggested reading

Biology and structure

- Moser & Jonkers 2025, 30 years of BRCA1: mechanistic insights and their impact on mutation carriers, Cancer Discovery — the one review to read first: what BRCA1 does, which domains do it, how variants are interpreted
- Khalizieva et al. 2025, BRCA1 and BRCA2: from cancer susceptibility to synthetic lethality, Genes & Development — companion review covering the clinical arc from risk to PARP inhibitors and resistance.
- Tarsounas & Sung 2020, The antitumorigenic roles of BRCA1–BARD1 in DNA repair and replication, Nat Rev Mol Cell Biol — mechanism at the level of domains and partners.
- Lou et al. 2014, Rapid evolution of BRCA1 and BRCA2 in humans and other primates, BMC Evolutionary Biology — highlight parts of BRCA1 that evolve fast, which is where alignment-based predictors get quiet.
- Findlay et al. 2018, Accurate classification of BRCA1 variants with saturation genome editing, Nature — 3,893 SNVs across 13 exons of the RING and BRCT regions in HAP1 cells; the single most important dataset for this project
- Starita et al. 2015, Massively parallel functional analysis of BRCA1 RING domain variants, Genetics — E3 ligase activity and BARD1 binding, variant by variant.
- Multiplexed DSB-repair assays of BRCT-domain variants, AJHG 2022 — a second, independent functional axis for the BRCT region.
- Functional assays for BRCA1 and BRCA2 variant analysis: new tools and their applications, npj Breast Cancer — recent review; use it to decide which assay measures what.

Cancer risk and clinical interpretation

- Kuchenbaecker et al. 2017, Risks of breast, ovarian and contralateral breast cancer for BRCA1 and BRCA2 mutation carriers, JAMA — the prospective cohort behind the penetrance figures everyone quotes: ~72% cumulative breast cancer risk to age 80 in BRCA1 carriers, strongly dependent on family history and variant position. Read it to understand what a "pathogenic" call actually commits a patient to.
- Fu et al. 2022, BRCA1 and breast cancer: molecular mechanisms and therapeutic strategies, Frontiers in Cell and Developmental Biology — the bridge from molecule to tumour: why BRCA1 loss produces predominantly triple-negative, early-onset, aggressive disease, and what that implies for treatment.
- ClinGen ENIGMA BRCA1/BRCA2 VCEP recommendations, AJHG 2024 — how evidence is actually weighed for these two genes.
- BRCA1 criteria specification registry (ClinGen) — the operational version of the same rules, including how functional data enter as PS3/BS3.

Computational methods

- Protein language models applied to variant effect (ESM-1v, ESM-2) and alignment-based generative models (EVE, GEMME). Read one of each and understand why they disagree. Check the https://proteingym.org/ leaderboard!
- AlphaMissense and REVEL as strong, ClinVar-trained baselines — and as a case study in circularity.
- SpliceAI / Pangolin for the RNA axis.
- MaveDB 2024, Genome Biology — what multiplexed assays exist and how their scores are normalised.
- CAGI7 "Annotate All Missense" challenge description — the submission format and evaluation logic this project imitates.
- Singh, Schmenger, Gonzalez-Sanchez et al. 2025, Discriminating activating, deactivating and resistance variants in protein kinases, Genome Medicine 17:133 (doi:10.1186/s13073-025-01564-z) — the closest published analogue of the mechanism half of this project: how sequence, structure and interaction evidence are combined to say not just whether a variant matters but what kind of change it makes.

13. Appendix: data for the mechanism question

There is no mechanism-labelled training set — not for BRCA1, not for any protein. What exists is one usable dataset per mechanism axis, measured on other proteins, plus a few BRCA1-specific resources that work as test material rather than as training material. The practical consequence is that a team trains one small model per axis and then arbitrates between them, which is what the taxonomy in section 3 is shaped for.

Axis	Dataset
Destabilisation	Tsuboyama et al. 2023 (Nature); ThermoMutDB, FireProtDB; Ssym and S669
Destabilisation vs. loss of function	VAMP-seq abundance assays (Matreyek et al. 2018: PTEN, TPMT)
Splicing	MFASS (Cheung et al. 2019); SpliceVarDB (Sullivan et al. 2024); SpliceAI, Pangolin; Findlay et al. 2018 (BRCA1)
Binding disruption	SKEMPI 2.0; IntAct; Interactome INSIDER
Zinc-site disruption	UniProt P38398; MetalPDB
Phosphorylation-site loss	PhosphoSitePlus, dbPTM, ActiveDriverDB; Johnson et al. 2023
Motifs in the disordered region	ELM
Integration and baselines	Mechismo, Mechnetor, Missense3D, ProtVar